Agile Product Portfolio Software Engineering Group Classwork

Group number: 10

Student ID	Student Name
3079411	Shuya yang
3002056	Xinyi Chen
2952821	Yiqun Liu

1. Planning (20%)

Conceptual model
Personas → Product Vision → User Stories → Features → Prototype → Tests

1.1 Personas

Persona Name	Role/User type	Goals	Pain Points	Related User Stories (IDs)
University Staff	University staff, living in single university accommodation, paying electricity individually	• Record daily electricity usage（monthly or weekly） and view personal usage history • Set a monthly electricity budget and track real-time consumption and remaining balance • Earn points through energy saving and redeem rewards • View personal energy data and monthly target progress	• No simple tool to record and check daily electricity usage • Difficulty tracking budget, leading to potential overspending • Lack of incentives to maintain long-term energy-saving habits • No unified platform for viewing personal energy records	S1 S2 S3
University Student	University student, living in shared student accommodation	• Record personal electricity usage and manage consumption behaviour • Set monthly electricity targets, progress, and savings points • Compare energy saving performance	• No clear way to view personal electricity data and trends • Lack of motivation systems for energy conservation • Cannot view personal energy-saving rank among peers	S1 S2 S3

		with others via leaderboards • Redeem points for rewards and check redemption history • Redeem prizes	• No unified system for reward redemption and history tracking	

Note: You need to have at least two personas. The personas should be used to justify the product vision and user stories.

1.2 Product Vision

For **university students and staff** who manage electricity costs either individually or through shared billing.

WHO need a simple, hardware-free web tool to log electricity usage, track monthly budgets, view history, earn energy-saving points, check leaderboards, and redeem rewards.

The **Green Campus Energy-Saving Platform (Ecopulse)** is a lightweight web-based platform
THAT supports:

- Daily electricity usage logging
- Monthly electricity budget and progress tracking
- Automatic points calculation for energy conservation
- Personal usage history and redemption history
- Energy-saving leaderboard display
- Points-based reward redemption

UNLIKE complex smart home systems that require hardware installation, or basic recording apps without incentives or ranking functions.

OUR PRODUCT is designed specifically for campus scenarios, free and easy to use, supports individual users, and uses points, rewards, and leaderboards to encourage long-term energy-saving behavior, fully meeting the campus users' complete needs of recording, managing, motivating, and redeeming.

Note: This product vision should be derived from the defined personas and guides backlog, prioritization, and design decisions.

1.3 Meeting Logs (Appendix)

2. Story Writing (20%)

2.1 Product Backlog and User Stories

Story ID	**Feature (epics)**	**Story Title**	**User Story**	**Acceptance Criteria**	**Priority (Must / Should / Could)**	**Justification of** prioritization/trade-off	**Story Owner**
S1	Electricity Logging & History	Log Daily Usage, View Monthly Records and Statistics	As a user, I want to log my daily electricity usage, have records saved automatically, view my full history, access current-month records, and see my monthly usage statistics, so that I can clearly understand my consumption behaviour.	Given I am logged into the platform, when I enter and submit daily electricity usage data, the system should save the record successfully. I should then be able to view all my historical records, filter them by the current month, and see the total usage for the month displayed.	Must	Core foundational function of the platform.	Xinyi Chen

S2	Budget & Points Tracking	Set Monthly Target, View Progress and Earn Energy Points	As a user, I want to set my monthly electricity target, view usage progress, see automatically calculated energy-saving rates, and earn points based on savings, so that I can control my consumption and receive incentives.	Given I am on the target management page, when I view or set my monthly electricity target, the system should display my remaining budget and usage progress. As I submit usage records, the system should automatically calculate and display my energy-saving rate and earned points based on the predefined rules.	Must	Directly addresses budget control and motivation pain points.	Shuyayang
S3	Redemption & Leaderboard	View Leaderboard, Redeem Rewards and Check History	As a user, I want to view the energy-saving leaderboard, redeem rewards with my points, and check my redemption history to stay motivated.	Given I am on the rewards platform, when I view the leaderboard page, I should see other users' energy-saving rankings. On the rewards page, my current available points should be displayed. If I select a reward and have sufficient points, the redemption should be allowed, and the record should appear in my redemption history.	Should	Improves user engagement and platform completeness.	Yiqun Liu

2.2 Sprint 1 Summary

Item	**Sprint 1** (Weeks 2-6) Discovery, MVP design and partial implementation	**Sprint 2** (Weeks 7-11) Refinement, testing & evaluation
Planned stories (Plan - P)	S1, S2	**S2,S3**
Delivered stories (Done - D)	S1	**S2,S3**
Stories not completed (No - N)	S2	
Key change planned for Sprint 2		Complete monthly points settlement, implement leaderboard, reward redemption, and redemption history viewing.

Note: Stories marked "P" or "N" should be reflected upon in sprint learning or retrospectives.

2.3 Group contribution and Individual contribution (Appendix)

3. Design (20%)

3.1 Design Models and Evidence

Diagram	Type (Class / Sequence)	Purpose	Related User Stories (Story IDs)	Design decision rationale
Sprint 1 Class Diagram	Class	Defines the core entities and their relationships to support login, energy record management, target setting, and weekly/monthly statistics.	S1,S2	Separated domain models into User, EnergyRecord, and EnergyService to decouple user management, data storage, and business logic. This design ensures clear data ownership (one-to-many relationship between User and EnergyRecord) and enables easy expansion for future features.
Sprint 1 Sequence Diagram	Sequence	Illustrates the step-by-step user interaction flow for login, setting monthly target, adding energy usage, and viewing weekly/monthly summaries.	S1,S2	Designed a layered flow (User → Frontend → Backend → DataStore) to clearly show API calls and data persistence. Each core function maps to a dedicated HTTP endpoint, ensuring separation of concerns and making it easy to implement and test each module independently.
Sprint 2 Class Diagram	Class	Extends the model with entities and services for points settlement, leaderboard, reward redemption, and redemption history tracking.	S2, S3	Introduced service classes (PointsService, LeaderboardService, RewardService) to encapsulate complex business logic, keeping the User class clean. Separated Reward and Redemption entities to clearly model available rewards and user transaction records, supporting audit and history features.

Sprint 2 Sequence Diagram	Sequence	Shows the interaction flow for automatic points settlement, viewing the leaderboard, redeeming rewards, and checking redemption history.	S2,S3	Continued the layered architecture from Sprint 1. Explicitly modeled the automatic points settlement process, the validation flow for reward redemption, and the data fetching flow for leaderboard and history. This ensures the full incentive loop (earn → compete → redeem) is transparent and implementable.

Note: You are **not** penalized for modifying or discarding models if justified by learning.

3.2 Stories and Evidence

For each story marked as "Done" in Sprint 1, it must be mapped to at least one test.

Story ID	Evidence Type (UML Diagram / Git Commit ID(s) / Test Evidence: test cases / their results)	Reference (file / commit / screenshot)	Acceptance Criteria Met (Yes / Partially / No)	Notes (if partially completed)
S1	**UML Diagram1**	**sprint1 class sprint1 sequence**	**Yes**	**Sequence diagrams and class diagrams**
S1	**Test Evidence**	ddac518	Yes	
S1	Git Commit ID(s)	ffc0d84, 26071d6, 8a82f63, aba915b, eb04596, dde36e7, c28a75d, 99af2a3, f113d79, a7ada7a 436717d ab16e16 245aafd eac0b71 f087a54 9d11aaf 0d06aa8 11b4dc5	Yes	
S2	**Test Evidence**	4375584, ddac518, ec90cb5，8bd8478，97541e8	Yes	
S2	Git Commit ID(s)	99af2a3, f113d79, a7ada7a, 8385e9d, b5b55ef, bf8d224, ce8696c, 7ac344a, 03b997a, 8805e11, 3b74084, eb04596, dde36e7, c28a75d, ef99ae4, d0ad939, 0462467, d586a09	Yes	
S2	**UML Diagram2**	**sprint1 class sprint1 sequence sprint2 class sprint2 SEQUNCE**	**Yes**	**Sequence diagrams and class diagrams**
S3	**UML Diagram3**	**sprint2 class sprint2 SEQUNCE**	**Yes**	**Sequence diagrams and class diagrams**
S3	**Test Evidence**	56dc918 d0254b3 44d28e1 f7568b4 7265556 00f1266 b683a47 e9f0a56 511e4bf cc9ccb4	**Yes**	

S3	Git Commit ID(s)	b52e5ac, 22bff57, 96984b0, 349c70b, ae5d311, 2998bdc, 908ccde 1568614 e5db4d4 eeee186 97541e8 8bd8478 e7b01bc 511e4bf cc9ccb4	**Yes**	

Note: If a story is marked "Done" without evidence, it **will not be treated as complete**.

4. Working Prototype and Implemented Features (20%)

Submit a **short video (3-5 minutes)** that demonstrates the implementation of 2 core end-to-end features and the refinement of the Sprint 1 feature. Each demonstrated feature must be traceable to user stories and acceptance criteria. The video should be submitted under the **"Product Video"** assignment on Canvas.

5. Testing and Evaluation (20%)

Story ID	Acceptance Criteria	Test Performed	Outcome (Pass / Partial / Fail)	Accessibility Checked	Notes (If partially tested)
S1	The user can log into the system and access the main page	Tested the /login endpoint with valid and invalid credentials, verifying responses for success, user not found, and incorrect password	Pass	Keyboard and Mouse Only	Login triggers demo data reset for testing purposes
S1	The user can input and save daily energy usage	Tested the /add-energy endpoint for both new entries and same-day updates, verifying that energy_records are correctly stored	Pass	Keyboard and Mouse Only	Supports overwriting energy data for the same day
S1	The user can view historical energy records	Tested the /user-data endpoint to retrieve records and accessed the /history page to verify correct display	Pass	Keyboard and Mouse Only	

S2	The user can set a monthly energy target	Tested the /set-target endpoint with valid input, non-numeric values, and negative values, verifying input validation and data update	Pass	Keyboard and Mouse Only	Includes input validation
S2	The system provides monthly energy statistics and prediction	Tested the /energy-summary endpoint, verifying correctness of fields such as total energy, usage progress, and prediction-related values	Pass	Keyboard and Mouse Only	Corresponds to Iteration 3 (prediction and points logic update)
S2	The system provides monthly energy statistics and prediction	Tested the /weekly-summary endpoint to verify weekly energy totals and progress calculation	Pass	Keyboard only	
S2	The system automatically settles energy-saving points	Verified that auto_settle_demo_data() is triggered during login and correctly updates points after a completed cycle	Pass		Corresponds to Iterations 14–15 (cycle settlement fixes)
S3	The user can view the leaderboard	Tested the /leaderboard-data endpoint, verifying sorting by total_points and correct rank assignment	Pass	Keyboard only	Corresponds to Iteration 1 (leaderboard API)
S3	The user can redeem rewards using points	Tested the /redeem-reward endpoint, including both successful redemption and insufficient points scenarios	Pass	Keyboard and Mouse Only	Corresponds to Iteration 5 (redemption logic refinement)
S3	The user can view redemption history	Tested the /redemption-history endpoint, verifying that records include reward name, points spent, and timestamp	Pass	Keyboard and Mouse Only	

Note: You may have more than one test per story. At least one test per story is required.

APPENDIX

A. Group Contribution

Story ID	Story Title	Predicted workload in hours	Actual workload in hours	Status (TODO/In progress/DONE)	Owner name
S1	Log Daily Electricity Usage	30	24	DONE	XINYI CHEN
S2	Log Daily Usage, View Monthly Records and Statistics	40	35	DONE	SHUYA YANG
S3	View Leaderboard, Redeem Rewards and Check History	34	25	DONE	YIQUN LIU
S4					
S5					

B. Individual Contribution

Team member	Role (s)	Stories Owned	Evidence (Git / Diagram / Test)	What could have been improved with more time?
Shuya yang	As the backend leader of the project, I independently completed the backend framework construction and backend code development. I implemented all core business logic interfaces including user login, electricity usage recording, monthly target setting, energy-saving statistics, points settlement, leaderboard calculation, and reward redemption to ensure stable system operation. Meanwhile, I independently designed all UML class diagrams and sequence diagrams for Iteration 1 and Iteration 2, and was responsible for project form filling, meeting arrangement, and user story (S2/S3) development, providing full support for project development and delivery.	S2,S3	**Backend**: 9a67bb2 9e20eda 8479ac5 52dd259 f87266f b52e5ac 22bff57 96984b0 349c70b ae5d311 ef99ae4 d0ad939 0462467 d586a09 2998bdc 908ccde 8805e11 3b74084 eb04596 dde36e7 c28a75d 99af2a3 f113d79 a7ada7a 8385e9d b5b555ef bf8d224 ce8696c 7ac344a 03b9997a ffc0d84 26071d6	Add unified authentication and permission control for all API interfaces to prevent unauthorized access.
Xinyi Chen	Served as the lead of persona and user story design, responsible for frontend-backend integration, test case design and execution, and page function logic	S1,S2	**Backend**: 4ac8c92 **Fronted:** 95efe34 0264578 d18baa9 3aacc10 f04d839 0578c30 e793bb5 518f7a1	Add color coding to the usage progress bar and leaderboard (e.g., over target / on target / energy-saving status) for

	development. Led the writing of user stories S1 and S2 and formulation of acceptance criteria; completed frontend-backend integration for login, homepage, history records, and monthly statistics modules; implemented user data acquisition, CRUD of electricity records, API debugging, and bug fixing. Designed and executed full-function test cases covering login, usage recording, target setting, points settlement, and redemption, verified availability and correctness of all APIs, and output test results and bug fixing records. Optimized frontend interaction logic, improved page routing and data display, and ensured complete and usable user flows.		e4d70b3 eeyad004 37e5f62 43890b9 4fd59b3 040a7cf 6c6622e e344dd6 5864f8b a41c9db 436717d ab16e16 **TEST：** e5db4d4 e7b01bc 511e4bf cc9ccb4 e9f0a56 7422816 50a00f7 4375584	at-a-glance status recognition.
Yiqun Liu	Responsible for frontend page development, CSS style design, UI beautification, interaction optimization, and project demo video recording. Completed UI layout and style beautification for login, dashboard, history, leaderboard, and redemption pages; optimized page structure, button interaction, modal styles, and responsive effects; upgraded the redemption interface visually by adding coupon image display area to improve readability and aesthetics. Implemented dynamic rendering and data display logic, optimized charts and status display; participated in improving user stories S2 and S3, ensured unified interface style and better user experience, and recorded and edited the project function demonstration video.	S2,S3	**Fronted:** eb65ff8 b32be27 1fcd53a 13cf02f 48ea6a2 f4d4942 6ddf89a c12d1ba 245aafd 9e7a60c a1449b7 **TEST：** eeee186 97541e8 8bd8478 56dc918 d0254b3 44d28e1 f7568b4 7265556	The redemption interface has undergone visual optimization and upgrades, with the addition of a coupon image display area. This combined text and image presentation visually showcases coupon information, enhancing page readability and ease of use, and improving the overall browsing experience.

C. Meeting Logs

You are required to schedule at least 2 meetings with your customer (TA).

Meeting ID	Team attendees	Customer (TA's name)	When	Agenda items	Actions and Owner
M1	Shuya yang Xinyi chen Yiqun liu	Aju Ani Justus	2.19 2:20-2:40PM	Update the TA on our current progress, select the most suitable user story, and get their feedback.	SHUYA YANG
M2	Xinyi chen Yiqun liu Shuya yang	Aju Ani Justus	2.24 3:40-4:00PM	Suggested to add another user story and prepare a	XINYI CHEN
M3	Xinyi chen Yiqun liu Shuya yang	Aju Ani Justus	3.19 3:40-4:00PM	Project progress reports and backend code demonstrations	YIQUN LIU

Meeting Diary for Team meetings				
Meeting ID	**Team attendees**	**When**	**Agenda items**	**Actions and Owner**
M1	Xinyi chen Yiqun liu Shuya yang	2.17 12:10-13:00	Discuss the initial project idea, define the core functional framework, identify optional features, and decide on the team roles.	YIQUN LIU
M2	Xinyi chen Yiqun liu Shuya yang	2.23 1:00-2:00PM	Review and finalize User Story 1 (electricity usage logging and history view), confirm acceptance criteria, and assign backend/frontend development tasks.	XINYI CHEN
M3	Xinyi chen Yiqun liu Shuya yang	4.10 12:22-15:20PM	Discuss User Story 2 (monthly target tracking and energy point calculation), refine the logic for energy-saving rate and points rules, and align the API design between backend and frontend.	SHUYA YANG

M4	Xinyi chen Yiqun liu Shuya yang	4.15 8:30-9:20AM	Review the progress of User Story 3 (leaderboard and reward redemption), confirm the implementation of the ranking algorithm, redemption process, and history records.	YIQUN LIU

Note: Meeting logs are not marked directly and are used only as supporting evidence.